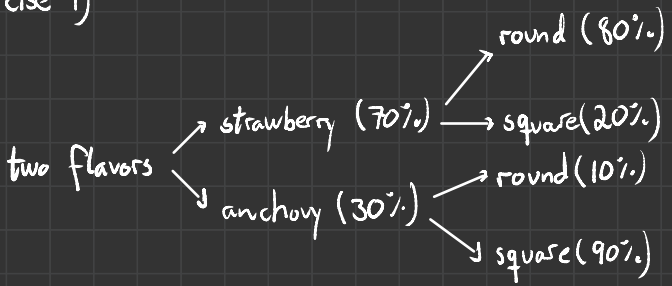
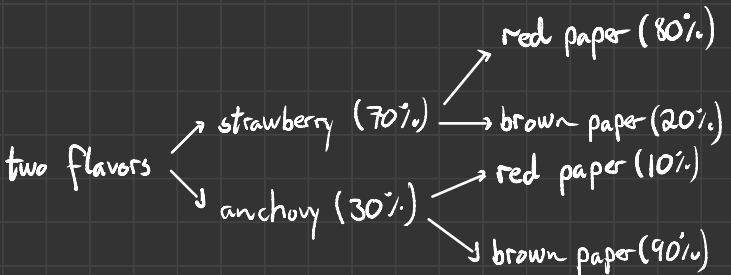
Exercise 1)





two flavors, strawberry or anchovy
two shapes, round or squared
two wrappers, red or brown

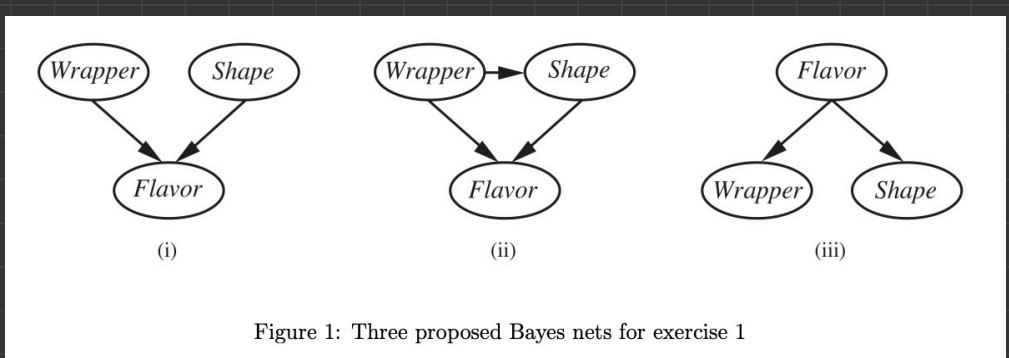


Figure 1: Three proposed Bayes nets for exercise 1

a)

i) Contradiction to the story. Diagram states that wrapper and shape decides flavor. This is not what's described.

ii) wrapper causes shape and flavor, shape causes flavor.
This is not what's described

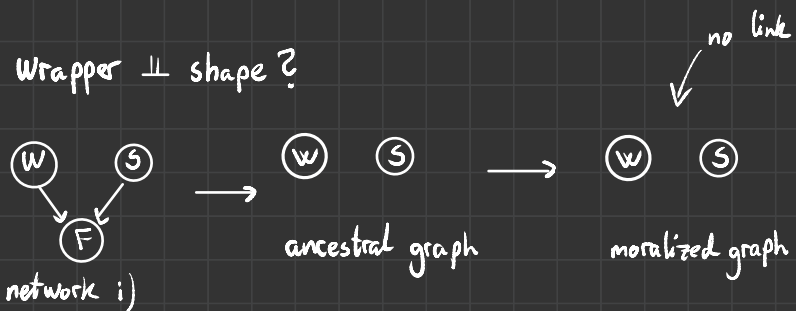
iii) this network represents the story, flavor decides both shape and wrapper

b) iii) is the best representation of the problem

P(Flavor)	1 parameter
P(Wrapper | Flavor)	2 parameters
P(Shape | Flavor)	2 parameters

Total 5 parameters

c) Wrapper ⫫ shape ?



Yes, wrapper is independent of shape

d) P(red paper) = P(red | strawberry) + P(red | anchovy)

know that 80% of strawberry are red and 70% are strawberry
know that 10% of anchovy are red and 30% are anchovy

P(red) = 0,7 · 0,8 + 0,3 · 0,1 = 0,56 + 0,03 = 59 = 59% of red paper

e) $P(\text{strawberry} \mid \text{round}, \text{red wrapper})$

$$P(\text{strawberry} \mid \text{round}, \text{red wrapper}) = \frac{P(\text{red wrapper} \mid \text{strawberry})\, P(\text{round} \mid \text{strawberry})\, P(\text{strawberry})}{\sum_f P(\text{red wrapper} \mid f)\, P(\text{round} \mid f)\, P(f)}$$

$$= \frac{P(\text{red wrapper} \mid \text{strawberry})\, P(\text{round} \mid \text{strawberry})\, P(\text{strawberry})}{P(\text{red wrapper} \mid \text{strawberry})\, P(\text{round} \mid \text{strawberry})\, P(\text{strawberry}) + P(\text{red wrapper} \mid \text{anchovy})\, P(\text{round} \mid \text{anchovy})\, P(\text{anchovy})}$$

$$= \frac{0{,}8 \cdot 0{,}8 \cdot 0{,}7}{0{,}8 \cdot 0{,}8 \cdot 0{,}7 + 0{,}1 \cdot 0{,}1 \cdot 0{,}3} = \frac{0{,}448}{0{,}448 + 0{,}003} = 0{,}993 \approx \underline{99{,}3\%}$$

If the candy has red wrapper and a round shape
it's a 99,3% of it's flavor is strawberry

f) unwrapped strawberry candy is worth s
unwrapped anchovy candy is worth a

What is expected value of unopened candy box?

Know that 70% are strawberry and 30% are anchovy hence the expected value for a box be

$$E[V] = 0{,}7s + 0{,}3a$$

Exercise 2)

Have utility function for money $U(x) = -e^{-\frac{x}{R}}$

a) Person Mary has utility function with $R = \$500$

Given choice of:

1) - receiving \$500
2) - lottery with 60% probability of winning \$5000, 40% of winning nothing

she acts rationally

Since Mary has a utility function with $R = \$500$ she is willing to risk \$500.

Apply the utility function

1) $U(500) = -e^{-\frac{500}{500}} = -e^{-1} = -0{,}368$

2) $\left.\begin{array}{l} U(5000) = -e^{-\frac{5000}{500}} = -e^{-10} \\ U(0) = -e^{0} = -1 \end{array}\right\} 0{,}6 \cdot e^{-10} + 0{,}4 \cdot (-1) \approx -0{,}4$

$-0{,}368 > -0{,}4$, so she will choose the first option

b) option 1) receive \$100 with certainty
option 2) 50% of winning \$500 and 50% of wiming nothing
Find R when the two options are indiffesent

we have $U(100) = -e^{-100/R}$

$\left.\begin{array}{l} U(500) = -e^{-500/R} \\ U(0) = -e^{0} = -1 \end{array}\right\} 0{,}5 \cdot -e^{-500/R} + 0{,}5 \cdot (-1)$

then we obtain $-e^{-100/R} = -\frac{1}{2}e^{-500/R} - \frac{1}{2}$ which is solved using the following .ipynb file and R is approximate 152.

```python
from scipy.optimize import brentq
import numpy as np

def f(R):
    return np.exp(-100/R) - 0.5*np.exp(-500/R) - 0.5

R_solution = brentq(f, 1, 10000)

print(f"{R_solution:.3g}")
```

[7] ✓ 0.0s

```
152
```